

# Generation Interconnection System Impact Study Report

## For Queue Project Q14-0326

Bonny Brook 138 kV

September 26, 2015

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0326. The Interconnection Customer proposes a 204.6 MW wind facility interconnecting at the Bonny Brook 138 kV substation in the Virginia area. The request was received on April 19, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 193 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,422,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                                 | Contingency                                  | Loading (%) | Rating (MVA) | DFAX (%) |
|----------------------------------------------------|----------------------------------------------|-------------|--------------|----------|
| Loisa Generation Facility Substation-N Anna 345 kV | Louisa Generation Facility-Carrabelle 345 kV | 106.8       | 1,686        | 5.0      |
| Sully Buttes-Yorktown 138 kV                       | Bonny Brook-Danc Lfgte Facility 138 kV Line  | 105.4       | 262          | 10.3     |
| Bixby Hill-Tombstone 500 kV Line                   | Sharyland-Tombstone 345 kV                   | 102.5       | 2,706        | 9.3      |

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

### 4. Required Network Upgrades and Cost Estimates

| Facility                                     | Required Upgrade           | Cost Estimate | Allocation to Project |
|----------------------------------------------|----------------------------|---------------|-----------------------|
| Louisa Generation Facility-North Anna 345 kV | Reconductor line           | \$20,394,000  | 1% (\$288,000)        |
| Sully Buttes-Yorktown 138 kV                 | Reconductor line           | \$1,016,000   | 18% (\$187,000)       |
| Bixby Hill-Tombstone 500 kV Line             | Reconductor line           | \$60,246,000  | 2% (\$947,000)        |
|                                              | Total allocated to project |               | \$1,422,000           |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.